

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XII: $\tilde{A}_2$ TOWERS — NO ABELIAN SECTOR, THE COVERING NORM THEOREM, AND THE TWO DIGRAPH TOWERS

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ABSTRACT. The specification asked for a two-dimensional congruence tower of $\tilde{A}_2$ complexes, a recursion for the Hecke critical groups along it, and a “rank-two Iwasawa law” $\mu_2 p^{2k} + \mu_1 p^k - \chi_2 k + \nu_0$ with $\chi_2 = 2$. Three things are established, and they change the framework. First, thick $\tilde{A}_2$ complexes have $b_1 = 0$ — the vanishing of first L^2 -cohomology for groups acting on thick $\tilde{A}_2$ buildings (Garland, Ballmann–Świątkowski) — and $H_1(Y; \mathbb{Z})$ is finite, computed exactly on our complexes $((\mathbb{Z}/2)^6, (\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14, (\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2)$: there are no $\mathbb{Z}_\ell$ -voltage towers in rank two, the commutative Iwasawa theory of [IV] has no rank-two sector, and every tower is a finite abelian chain or a congruence tower. Second, along the finite abelian chains $X_3 \leftarrow Y_{21} \leftarrow Y_{84} \leftarrow Y_{168}$ (and through Y_{24}, Y_{42}) the covering pushforwards intertwine A_1, A_2 and the Hecke–Laplacian, are surjective on the Hecke critical groups, and their kernel orders obey the rank-two character mechanism $|\text{Jac}^H(Y_{k+1})|/|\text{Jac}^H(Y_k)| = \frac{n_k}{n_{k+1}} \prod_{\text{new } j} |P_j(1)|$, a product of twisted GL_3 local L -values, verified to 10^{-6} on five coverings together with its Hodge analogue. Third, the two natural digraph towers of an $\tilde{A}_2$ complex — the geodesic edge tower E_k (q^2 -regular, $n_k = n(q^2 + q + 1)q^{2(k-1)}$, the $(k, 0)$ -coset counts of PGL_3) and the chamber tower F_k (q -regular) — obey the line-digraph recursion of [III] with degree q^2 and q , giving the exact laws $\text{ord}_p \kappa(E_k) = \frac{2n(q^2 + q + 1)}{q^2} p^{2k} - 2k + \nu$ and $\text{ord}_p \kappa(F_k) = \frac{3|F|}{q} p^k - k + \nu'$: the “higher Euler term” $\chi_2 = 2$ is $\text{ord}_p(q^2)$, the valuation of the regularity degree, one exponential term per tower and no mixed law. The sandpile of the geodesic edge digraph factors as an Eisenstein term, the product $\prod_j q^6 P_j(q^{-2})$ of GL_3 local L -values at $u = q^{-2}$, and the parahoric factor, by the edge-flow theorem of [XI]. A two-dimensional degeneracy calculus for the incidence matrices (S, T, M) and the three face maps d_i is proved, and it yields a combinatorial proof that A_1 and A_2 commute. The growth law of Jac^H along a genuine infinite congruence tower is left open, with the reason stated: no computable second level exists, and no abelian shortcut exists in principle.

STATUS OF THE RESULTS

Every numbered statement is proved except where marked “numeric”. The classical inputs are the vanishing theorem for the first L^2 -cohomology of groups acting on thick $\tilde{A}_2$ buildings [2, 1] (quoted; our complexes’ Betti numbers are also computed exactly), the line-digraph recursion [III, Thm. 2.1], the edge-flow theorem [XI, Thm. 3.3], and the norm-surjectivity argument of [IV, Lem. 1.3]. All Smith forms are p -adic with the prime sets certified as in [X, §6] or by determinants of maximal minors (Section 1). Section 6 records what is open and the corrections to the specification.

Date: 19 September 2026.

2020 Mathematics Subject Classification. Primary 11R23, 20E42; Secondary 05E45, 05C50, 11F70, 20F65.

Key words and phrases. $\tilde{A}_2$ complex, property (T), abelian cover, Hecke critical group, covering norm, line digraph, sandpile recursion, Iwasawa growth, degeneracy maps.

Paper XII of the series; Papers I–XI are cited as [I]–[XI]. The verification script `pgl3_tower.py` in the ancillary bundle reproduces every computational claim; its output is archived as `pgl3_tower_output.txt`.

1. NO ABELIAN SECTOR IN RANK TWO

Theorem 1.1 (No abelian towers). *Let $Y = \Gamma \backslash B$ be a finite quotient of a thick $\tilde{A}_2$ building of order $q \geq 2$ by a group acting freely. Then $H^1(\Gamma; \mathbb{Q}) = H^1(Y; \mathbb{Q}) = 0$, i.e. $b_1(Y) = 0$, and $H_1(Y; \mathbb{Z})$ is finite. Consequently Y has no connected $\mathbb{Z}_\ell$ -voltage covers for any prime ℓ : every abelian cover of Y is a finite cover with group a quotient of the finite group $H_1(Y; \mathbb{Z})$, and the limit modules X_∞ of [IV] have no rank-two analogue.*

Proof. Γ acts properly and cocompactly on B ; by Garland's vanishing theorem as extended to all $q \geq 2$ by Ballmann–Świątkowski [2, 1], $H^1(\Gamma; \mathbb{R}) = 0$ (indeed Γ has property (T)). Since Y is aspherical with $\pi_1(Y) = \Gamma$, $b_1(Y) = 0$ and $H_1(Y; \mathbb{Z}) = \Gamma^{\text{ab}}$ is finite. A voltage cover with group $\mathbb{Z}/\ell^k$ is classified by a class in $H^1(Y; \mathbb{Z}/\ell^k) = \text{Hom}(H_1(Y; \mathbb{Z}), \mathbb{Z}/\ell^k)$, whose image has bounded order; a $\mathbb{Z}_\ell$ -tower would need unbounded ones. On Y_{21}, Y_{24}, Y_{42} the groups $H_1(Y; \mathbb{Z})$ are $(\mathbb{Z}/2)^6$, $(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$, $(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$ (exact: p -adic Smith forms of the coordinate matrix of ∂_2 in a basis of $\ker \partial_1$, with the prime set certified by the gcd of maximal minors, 64, 448, 256); on Y_{84}, Y_{168} , $b_1 = 0$ by exact ranks (Table 1, line (B)). $\square$

Remark 1.2 (What this changes). In rank one every finite graph has $b_1 = r \geq 2$ and carries $\mathbb{Z}_\ell$ -voltage towers for every ℓ ; Papers II, IV, VII and IX built the commutative Iwasawa theory on them, and the Iwahori tower of [III], [V], [VII], [VIII] was the non-normal exception. In rank two the exception is the rule: the only infinite towers of $\tilde{A}_2$ complexes are congruence-type towers with non-abelian (indeed property-(T)) deck structure, and the only abelian structure available is the finite group $H_1(Y; \mathbb{Z})$. The specification's "rank-two Iwasawa theory" is therefore, from the start, the non-commutative theory that [V] and [VII] asked for in rank one; nothing commutative can be borrowed.

2. THE FINITE ABELIAN CHAINS AND THE COVERING NORM THEOREM

The complexes of [X] are abelian covers of the three-vertex type quotient X_3 of the Cartwright–Steger-type group with deck group $H_1(X_3; \mathbb{Z}) = \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/14$; every quotient of that group gives a cover, and quotient maps give covering maps π between covers. We use the chains

$$X_3 \leftarrow Y_{21} \leftarrow Y_{84} \leftarrow Y_{168}, \quad Y_{21} \leftarrow Y_{42} \leftarrow Y_{168}, \quad Y_{24} \leftarrow Y_{168},$$

of degrees 7, 4, 2; 2, 4; 7. Write $P_\pi \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ for the vertex pushforward (sum over fibres), $A_i^{(k)}$ for the Hecke operators of Y_k , and

$$\Delta_k^H = (q^3 - 1)I + A_1^{(k)} - qA_2^{(k)}.$$

Theorem 2.1 (Covering norm theorem). *For a covering $\pi: Y_{k+1} \rightarrow Y_k$ of $\tilde{A}_2$ complexes:*

- (i) $P_\pi A_i^{(k+1)} = A_i^{(k)} P_\pi$ for $i = 1, 2$, and hence $P_\pi \Delta_{k+1}^H = \Delta_k^H P_\pi$;
- (ii) *the induced norm π_* from $\text{coker } \Delta_{k+1}^H$ to $\text{coker } \Delta_k^H$ is surjective, preserves the degree functional, and restricts to a surjection of $\text{Jac}^H(Y_{k+1})$ onto $\text{Jac}^H(Y_k)$ whose kernel has order $|\text{Jac}^H(Y_{k+1})|/|\text{Jac}^H(Y_k)|$;*
- (iii) *(character mechanism) that order equals $\frac{n_k}{n_{k+1}} \prod_{j \text{ new}} |P_j(1)|$, the product over the joint eigenpairs of $(A_1^{(k+1)}, A_2^{(k+1)})$ not lifted from Y_k , i.e. over the nontrivial characters χ of the deck group and the joint eigenpairs of the χ -twisted operators $(A_1(\chi), A_2(\chi))$ on Y_k : a product of twisted GL_3 local L -values at $u = 1$. The same holds for the Hodge critical groups with $|P_j(1)|$ replaced by the Laplacian eigenvalues $2(q^2 + q + 1) - 2\text{Re } a_j$.*

Proof. (i) A covering is bijective on in- and out-arcs at every vertex, so fibre sums commute with the type-raising and type-lowering adjacencies; Δ^H is a polynomial in them. (ii) The argument of [IV, Lem. 1.3] applies verbatim: P_π is surjective, $\mathbf{1}^t \Delta^H = 0$ makes $\deg$ well defined, $\deg \circ \pi_* = \deg$, and

Tor coker $\Delta^H = \ker \deg$ by [X, Thm. 3.2(ii)]; surjectivity on torsion follows, and the kernel order is forced by counting. (iii) The deck group acts on Y_{k+1} commuting with A_1, A_2 , so the joint spectrum of $(A_1^{(k+1)}, A_2^{(k+1)})$ is the union over characters χ of the joint spectra of the twisted operators on Y_k ; the trivial character gives the spectrum of Y_k . Apply [X, Thm. 3.2(ii)] at both levels and divide. Verified numerically to 10^{-6} on the five coverings, together with the exact integrality of the ratios and the exact intertwining identities (Table 1, line (A)). $\square$

Data. $|\text{Jac}^H|$ along the chains, with the 2-adic valuations: Y_{21} : $3 \cdot 7^{19}$ (2^0); Y_{42} : $2^2 \cdot \text{odd}$; Y_{24} : $2^{18} \cdot \text{odd}$; Y_{84} : $2^7 \cdot \text{odd}$; Y_{168} : $2^{18} \cdot \text{odd}$. The groups: $\text{Jac}^H(Y_{84}) = (\mathbb{Z}/7)^{18} \oplus (\mathbb{Z}/299593)^5 \oplus \mathbb{Z}/2097151 \oplus \mathbb{Z}/29360114 \oplus \mathbb{Z}/117440456 \oplus \mathbb{Z}/352321368$ (new here); the others are in [X]. Kernel orders of the norms have 2-adic valuations 11 ($Y_{168} \rightarrow Y_{84}$), 7 ($Y_{84} \rightarrow Y_{21}$), 0 ($Y_{168} \rightarrow Y_{24}$), 16 ($Y_{168} \rightarrow Y_{42}$), 2 ($Y_{42} \rightarrow Y_{21}$). There is no monotone p -adic law along these finite chains, and none is to be expected: a finite abelian chain is not a tower.

3. TWO-DIMENSIONAL DEGENERACY CALCULUS

Let $S, T, M \in M_{n \times |E|}(\mathbb{Z})$ be the tail, head and third-vertex incidence matrices of [XI, §3], and $d_0, d_1, d_2 \in M_{|E| \times |F|}(\mathbb{Z})$ the three face maps sending a chamber (a, b, c) (types 0, 1, 2) to its typed edges $(a \rightarrow b)$, $(b \rightarrow c)$, $(c \rightarrow a)$.

Theorem 3.1 (Incidence calculus in rank two). *For an $\tilde{A}_2$ complex satisfying [X, Def. 1.1(i)–(iv)] with $\text{PG}(2, q)$ links:*

$$\begin{aligned} ST^t &= A_1, & TS^t &= A_2, & SS^t &= TT^t = (q^2 + q + 1)I, \\ SM^t &= (q + 1)A_2, & TM^t &= (q + 1)A_1, \\ MM^t &= (q^2 + q + 1)(q + 1)I + A_{(1,1)}, \\ A_{(1,1)} &:= A_1A_2 - (q^2 + q + 1)I = A_2A_1 - (q^2 + q + 1)I, \\ \partial_1 &= T - S, & \Delta_0 &= (T - S)(T - S)^t = 2(q^2 + q + 1)I - A_1 - A_2, \\ \partial_2 &= d_0 + d_1 + d_2, \\ \sum_i d_i d_i^t &= (q + 1)I_E, & d_i^t d_j &= 0 \ (i \neq j), \\ K &:= \sum_i d_i d_{i+1}^t, & L_E &= T^t S - K. \end{aligned}$$

In particular $A_1A_2 = A_2A_1$: the two Hecke operators commute, by the symmetry of MM^t .

Proof. The first line is [XI, proof of Thm. 3.3] and Lemma [XI, 3.2(a)]: $(SM^t)(x, w)$ counts edges $x \rightarrow y$ whose chamber contains w , i.e. chambers on the edge $\{x, w\}$ with w a type-raising-twice neighbour, $q + 1$ of them; similarly TM^t . For MM^t : the diagonal counts chambers through v , $(q^2 + q + 1)(q + 1)$; for $v \neq w$, $(MM^t)(v, w)$ counts edges $x \rightarrow y$ with $\{x, y, v\}$ and $\{x, y, w\}$ chambers, so x is a common out-neighbour of v, w and y a common in-neighbour; given the common out-neighbour x , in the link of x the vertices v, w are two distinct points and y must be a line through both — exactly one — so $(MM^t)(v, w) = \#\{x: v \rightarrow x, w \rightarrow x\} = (A_1A_2)(v, w)$ for $v \neq w$, and the diagonal of A_1A_2 is $q^2 + q + 1$. Since MM^t is symmetric, $A_1A_2 = (A_1A_2)^t = A_2A_1$. The Laplacian line is $\partial_1 = T - S$ expanded with the first line. The face maps: every chamber has one edge of each type, so $\sum_i d_i d_i^t$ is the diagonal matrix of chambers per edge, $q + 1$; $d_i^t d_j = 0$ for $i \neq j$ because an edge has one type; $K(e, e')$ counts chambers in which e, e' are consecutive, which is the chamber-continuation term subtracted in [XI, Def. 3.1]. Verified exactly on Y_{168} (Table 1, line (D)). $\square$

Remark 3.2 (The three degeneracy operators). In rank one, [V] had two degeneracy maps τ, η with $\tau_* \eta^* = A_k$. In rank two the three face maps $d_i: C_2 \rightarrow C_1$ together with the two vertex maps S, T generate both Hecke operators, their commutation, the Hodge Laplacian Δ_0 , the chamber

continuation K and the edge flow L_E : the whole vertex- and edge-level Hecke theory of the complex is a polynomial identity in (S, T, M, d_0, d_1, d_2) , and the identity $A_1 A_2 = A_2 A_1$ — normally quoted from the commutativity of the spherical Hecke algebra — is here a count of lines through two points.

4. THE TWO DIGRAPH TOWERS

Definition 4.1 (Edge and chamber towers). For an $\tilde{A}_2$ complex Y with the hypotheses of Theorem 3.1 let $E_1 := (E, L_E)$ be the geodesic edge digraph and $F_1 := (F^\pm, L_F)$ the chamber digraph of [XI, Rem. 3.6] (oriented chambers $(a, b, c) \rightarrow (b, c, d)$, $d \neq a$), and for $k \geq 1$ let $E_{k+1} := L(E_k)$, $F_{k+1} := L(F_k)$ be their line digraphs. Vertices of E_k are geodesic type-raising galleries of k edges; their number is $n_k(E) = n(q^2 + q + 1)q^{2(k-1)}$, the number of vertices at coweight distance $(k, 0)$ from a vertex of the building, i.e. the index of the PGL_3 congruence subgroup fixing a $(k, 0)$ -flag: E_k is the rank-two Iwahori path tower in the $(k, 0)$ direction.

Theorem 4.2 (Recursions and growth laws). E_1 is Eulerian q^2 -regular and strongly connected, F_1 is Eulerian q -regular and strongly connected (exact on Y_{168} ; in general by [XI, Lem. 3.2(a)] and its chamber analogue). Hence, by [III, Thm. 2.1] applied with $d = q^2$ and $d = q$, for all $k \geq 1$

$$\kappa(E_{k+1}) = (q^2)^{n_k(E)(q^2-1)-1} \kappa(E_k), \quad \kappa(F_{k+1}) = q^{n_k(F)(q-1)-1} \kappa(F_k),$$

and for $q = p$ the exact growth laws

$$\mathrm{ord}_p \kappa(E_k) = \frac{2n(q^2 + q + 1)}{q^2} p^{2k} - 2k + \nu_E, \quad \mathrm{ord}_p \kappa(F_k) = \frac{3|F|}{q} p^k - k + \nu_F,$$

with constants fixed by $\kappa(E_1)$, $\kappa(F_1)$. The nonzero spectra of A_{E_k} and of A_{F_k} are independent of k (spectral blindness), so the zeta function of E_k is $\det(I - uL_E)^{-1}$ at every level and contains the whole nontrivial GL_3 Hecke spectrum [XI, Thm. 3.3].

Proof. Regularity: each typed edge has q^2 continuations and q^2 predecessors ([XI, Lem. 3.2(a)] and its reverse); each oriented chamber has q successors and q predecessors (the $q + 1$ chambers on an edge minus the one you came from). Line digraphs of Eulerian d -regular strongly connected digraphs are again such, on d times as many vertices, with $\det(zI - A_{L(G)}) = z^{m-n} \det(zI - A_G)$ [III, eq. (2.1)]; the recursion is [III, Thm. 2.1]; summing, $\sum_{j < k} 2(n_j(E)(q^2 - 1) - 1) = 2n_1(E)(q^{2(k-1)} - 1) - 2(k-1)$ gives the edge law and likewise the chamber law. Sanity check of [III, Thm. 2.1] with $d = 4$ on the complete digraph K_5 : $\kappa = 125$, $\kappa(L) = 4^{14} \cdot 125$ (Table 1, line (L)). $\square$

Remark 4.3 (The higher Euler term). The specification predicted $\chi_2 = 2$. It is right, and for a transparent reason: in [IX, Thm. 4.1(iv)] the Euler term of the rank-one Iwahori tower was identified as $-\mathrm{ord}_p n_k$, the valuation of the vertex count in the residue normalization $\prod_{\mu \neq 0} \mu = n_k \kappa_k$; here $n_k(E) = n_1 q^{2(k-1)}$ has $\mathrm{ord}_p n_k = 2k + \mathrm{const}$, so the Euler term is $-2k$, and for the chamber tower $-k$. In general the Euler term of a line-digraph tower of degree d is $-\mathrm{ord}_p(d) \cdot k$: one Eisenstein direction removed per level, weighted by the p -adic size of the regularity degree. There is no mixed law $\mu_2 p^{2k} + \mu_1 p^k$: each tower has one exponential term, with μ the arc density n_1/d times $\mathrm{ord}_p d$.

Proposition 4.4 (The sandpile of the geodesic edge digraph). With the notation of [XI, Cor. 3.5],

$$\kappa(E_1) = \frac{3q^4}{n_1} \cdot \prod_{j \text{ nontrivial}} q^6 P_j(q^{-2}) \cdot \prod_{\lambda \text{ parahoric}} (q^2 - \lambda),$$

where $q^6 P_j(q^{-2}) = \prod_i (q^2 - \lambda_{j,i})$ is, up to the factor q^6 , the inverse local L -value of the j -th eigenform at $u = q^{-2}$, and the parahoric factor runs over the $4n + 6$ eigenvalues of L_E of modulus $q^{1/2}$. On Y_{168} (numeric): $\log \kappa(E_1) = 1623.9567$, decomposed as $\log(3q^4/n_1) = -3.1987$, pencil part 684.7469, parahoric part 942.4085, which add up to 1623.9567.

Proof. E_1 is Eulerian and strongly connected, so [III, eq. (2.2)] gives $n_1\kappa(E_1) = \prod_{\mu \neq 0} \mu$ over the nonzero eigenvalues of $q^2I - L_E$, i.e. over all eigenvalues $\lambda \neq q^2$ of L_E . By [XI, Cor. 3.5] these are: the two rotations $q^2\omega, q^2\omega^2$, contributing $(q^2 - q^2\omega)(q^2 - q^2\omega^2) = 3q^4$; the three roots of each nontrivial pencil P_j ; and the parahoric eigenvalues. Exact $\kappa(E_1)$ (a 1176×1176 determinant of size e^{1624}) is not computed. $\square$

5. THE VERIFICATION BATTERY

check	complexes	result
(B) $b_1 = 0$; $H_1(Y; \mathbb{Z}) = (\mathbb{Z}/2)^6$, $(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$, $(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$; prime sets certified by minor gcds 64, 448, 256	Y_{21}, Y_{24}, Y_{42} (b_1 also Y_{84}, Y_{168})	exact
(A) $P_\pi A_i = A_i P_\pi$, $P_\pi \Delta^H = \Delta^H P_\pi$; Jac^H at five levels; ratios integral; character mechanism to 10^{-6} ; Hodge analogue	five coverings	exact / numeric
(D) the fourteen identities of Theorem 3.1, including $A_1 A_2 = A_2 A_1$	Y_{168}	exact
(E) E_1 : q^2 -regular in and out, strongly connected, $n_1 = 1176$; spectral accounting of $\kappa(E_1)$; constants $\mu_2 = 588$, $\chi_2 = 2$	Y_{168}	exact / numeric
(F) F_1 : q -regular in and out, 3528 vertices; constants $\mu_1 = 1764$, $\chi = 1$	Y_{168}	exact
(L) [III, Thm. 2.1] with $d = 4$ on K_5 : $\kappa(L) = 4^{14}\kappa$	—	exact

TABLE 1. Verification (`pgl3_tower.py`).

6. SCOPE, RESIDUE, CORRECTIONS

Remark 6.1 (Solved, open, impossible). *Solved*: the non-existence of abelian towers in rank two and its consequence for the series’ framework; the covering norm theorem with the rank-two character mechanism on the finite abelian chains; the two-dimensional incidence calculus with the combinatorial commutation of A_1, A_2 ; the edge and chamber digraph towers with their exact recursions and growth laws and the identification of χ_2 ; the L -value structure of the sandpile of the geodesic edge digraph. *Open*: (a) the growth law of Jac^H along a genuine infinite congruence tower $\Gamma \supset \Gamma_1 \supset \dots$ of $\tilde{A}_2$ complexes — no second level is computable here (the maximal abelian cover of Y_{168} would have $168 \cdot |H_1(Y_{168})|$ vertices), and Theorem 1.1 shows no abelian shortcut exists; (b) an exact value of $\kappa(E_1)$ on Y_{168} and the p -adic constant ν_E ; (c) a closed form for the parahoric factor of Proposition 4.4, which should be Kang–Li’s chamber factor at $u = q^{-2}$; (d) the pro-module of an infinite congruence tower — which, by Theorem 1.1, is a module over a non-commutative completed Hecke algebra from the start, the object [V, Prob. 5.1] asked for in rank one. *Impossible or false as stated*: a $\mathbb{Z}_p$ -tower of $\tilde{A}_2$ complexes; a two-term law $\mu_2 p^{2k} + \mu_1 p^k$ for a single line-digraph tower.

Remark 6.2 (Corrections to the task specification). (1) “ $X_\infty^{(2)} = \varprojlim (\text{Jac}^H(Y_k) \otimes \mathbb{Z}_p)$ ” over an abelian tower does not exist: there are no abelian towers (Theorem 1.1); over the finite chains the inverse limit is just the top group. (2) The proposed recursion $|\text{Jac}^H(Y_{k+1})| = q^{f(n_k, E_k, F_k)} |\text{Jac}^H(Y_k)|$ is not what happens along coverings of $\tilde{A}_2$ complexes: the ratio is a product of twisted L -values (Theorem 2.1(iii)), not a pure power of q (its 2-adic valuations along our chains are 11, 7, 0, 16, 2); a pure-power recursion is what the *digraph* towers obey (Theorem 4.2), for the ordinary sandpile groups, not for Jac^H . (3) The law $\mu_2 p^{2k} + \mu_1 p^k - \chi_2 k + \nu_0$ splits into two separate one-term laws, one per tower; $\chi_2 = 2$ is $\text{ord}_p(q^2)$ (Remark 4.3). (4) The “2D line complex” $\mathcal{L}_2(Y_0)$ is a digraph, not a two-dimensional complex; the levels E_k carry no A_2 , and the Hecke–Laplacian Δ_k^H is not defined on them — the Hecke data live at level one, in $\det(I - uL_E)$, and are constant along the tower.

ACKNOWLEDGEMENT OF PROVENANCE AND CAVEATS

Unrefereed preprint. Every numbered statement is proved from Papers I–XI and finite combinatorics, except the vanishing of b_1 for general thick quotients, which is quoted from [2, 1] and independently computed on our five complexes. Spectral statements marked numeric are floating point. Bibliographic data of [1] were checked against the publisher page and those of [2] against the JSTOR record.

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